

Computer and Network Security — Lecture 06

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Brief Review of Modular Arithmetic (Con't)

Why Modular Arithmetic in Cryptography?

Modular arithmetic provides the mathematical foundation for many cryptographic algorithms:

- ① Cryptography operates on finite fields as they offer specific mathematical properties that can be used to simplify large computations.
- ② Modular arithmetic ensures that the results of cryptographic operations stay within a finite field (in groups, rings, and fields).
- ③ Cryptography requires hard problems. Some problems become hard with modular arithmetic (e.g., DLP, PFP, ECDLP, ...).
- ④ Cryptography is implemented digitally. If you work with modular arithmetic, you have guarantees about the largest value you will see and can allocate the correct amount of space to hold values.

Fun Fact. You are using modular arithmetic everyday.

The Modulo Operator: Congruence

Let $a, b, m \in \mathbb{Z}$, and $m > 0$ (called, the modulus). We write:

$$\forall a, b \in \mathbb{Z} : a \equiv b[m] \text{ simply means that } \exists k \in \mathbb{Z} : a = k \cdot m + b$$

(b is called the residue, or remainder)

Or

$$\forall a, b \in \mathbb{Z} : a \equiv b[m] \text{ simply means that } m \mid (a - b)$$

(m divides $(a - b)$)

We would say that: a is congruent to b modulo m .

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Example. Compute $15[12]$, $33[10]$, and $23[5]$.

- $15 \equiv 3 [12]$
- $33 \equiv 3 [10]$
- $23 \equiv 3 [5]$

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How many remainders are there if $m = 9$ and $a = 42$?

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We would say that: a is congruent to b modulo m .

There exists multiple remainders (the remainder is not unique).

So, how many remainders are there if $m = 9$ and $a = 42$?

The Modulo Operator: Congruence

Let $a, b, m \in \mathbb{Z}$, and $m > 0$ (called, the modulus). We write:

Consider $a = 42$ and $m = 9$, s.t., $a = k \cdot m + b$ in $\mathbb{Z}$:

- $42 = 4 \cdot 9 + 6$, here the remainder is $b = 6$ (9 divides 42-6)
- $42 = 3 \cdot 9 + 15$, here the remainder is $b = 15$ (9 divides 42-15)
- $42 = 2 \cdot 9 + 24$, here the remainder is $b = 24$ (9 divides 42-24)
- $42 = 1 \cdot 9 + 33$, here the remainder is $b = 33$ (9 divides 42-33)
- $42 = 5 \cdot 9 + (-3)$, here the remainder is $b = -3$ (9 divides 42+3)
- $42 = 6 \cdot 9 + (-12)$, here the remainder is $b = -12$ (9 divides 42+12)
- ...

We can claim that the remainder is not unique as long as the standard “non-negative restriction” is not enforced, and that the remainder should not necessarily be within the range of $(0 \text{ to } m - 1)$.

Equivalence (Residue) Classes

Residue classes are sets of infinite integers that produce the same remainder when divided by a modulus m :

$$\bar{a}_m = \{x \in \mathbb{Z} \mid x \equiv a[m]\}, \text{ where } m \in \mathbb{Z}_m$$

Example. In modulo $m = 5$, the residue classes are:

$$\bar{0}_m = \{\dots, -10, -5, 0, 5, 10, \dots\},$$

$$\bar{1}_m = \{\dots, -9, -4, 1, 6, 11, \dots\},$$

$$\bar{2}_m = \{\dots, -8, -3, 2, 7, 12, \dots\},$$

$$\bar{3}_m = \{\dots, -7, -2, 3, 8, 13, \dots\},$$

$$\bar{4}_m = \{\dots, -6, -1, 4, 9, 14, \dots\},$$

All members of the class behave equivalently modulo m .

If we had to compute $(13 \cdot 16 - 8)[5]$, we simply compute

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Important application is to compute, e.g., $3^8[7] \equiv ?$.

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If we had to compute $(13 \cdot 16 - 8)[5]$, we simply compute $(3 \cdot 1 - 3)[5]$.

Important application is to compute, e.g., $3223^{800}[322] \equiv ?$.

Modular Arithmetics Properties

Here are some modular arithmetic properties you may use:

- ① $(a[m] + b[m])[m] = (a + b)[m]$.
- ② $(a[m] - b[m])[m] = (a - b)[m]$.
- ③ $(a[m] \cdot b[m])[m] = (a \cdot b)[m]$.
- ④ $(a + b)[m] = (b + a)[m]$.
- ⑤ $(a \cdot b)[m] = (b \cdot a)[m]$.
- ⑥ $((a + b) + c)[m] = (a + (b + c))[m]$
- ⑦ $((a \cdot b) \cdot c)[m] = (a \cdot (b \cdot c))[m]$
- ⑧ $(0 + a)[m] = a[m]$
- ⑨ $(1 \cdot a)[m] = a[m]$
- ⑩ $(a^p)[m] = ((a[m])^p)[m]$
- ⑪ $\forall a \in \mathbb{Z}_m, \exists! b \in \mathbb{Z}_m \text{ s.t. } a + b \equiv 0[m]$
- ⑫ $\forall a \in \mathbb{Z}_m - \{0\}, \text{ If } \gcd(a, m) = 1, \text{ then } \exists! a^{-1} \in \mathbb{Z}_m, a \cdot a^{-1} \equiv 1[m]$

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Modular Arithmetics Properties

Here are some modular arithmetic properties you may use:

① Extended Euclidean Algorithm (EEA):

② To find the inverse a^{-1} of a modulo m :

- ③ ① $(A_1, A_2, A_3) = (1, 0, m)$
- ④ ② $(B_1, B_2, B_3) = (0, 1, a)$
- ⑤ ③ If $(B_3 == 0)$ then a^{-1} does not exist (cuz $a == 0$).
- ⑥ ④ If $(B_3 == 1)$ then $B_2 = a^{-1}[m]$.
- ⑦ ⑤ $Q = [A_3 \div B_3]$
- ⑧ ⑥ $(T_1, T_2, T_3) = (A_1 - Q \cdot B_1, A_2 - Q \cdot B_2, A_3 - Q \cdot B_3)$
- ⑨ ⑦ $(A_1, A_2, A_3) = (B_1, B_2, B_3)$.
- 10 ⑧ $(B_1, B_2, B_3) = (T_1, T_2, T_3)$.
- 11 ⑨ Goto 3.

12 $(a^{-1})[m] = ((a[m])^{-1})[m]$

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12 ➡ What is the inverse a^{-1} of $a = 17$ modulo 43?

Modular Arithmetics Properties

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It is $a^{-1} = 38$

Modular Arithmetics Properties

Here are some modular arithmetic properties you may use:

1 Extended Euclidean Algorithm (EEA):

2 Applying the Euclidean algorithm:

3 $43 = 17 \times 02 + 09$

4 $17 = 09 \times 01 + 08$

5 $09 = 08 \times 01 + 01$ (so, $\gcd(43, 17)=1$)

6 $08 = 08 \times 01 + 00$

7 Then, by applying EEA: (we want $1 = (43 \times k) + (17 \times a^{-1})$)

8 $01 = 09 - 08 \times 01$

9 $01 = 09 - (17 - 09 \times 01)$

10 $01 = 09 - 17 + 09$

11 $01 = 09 \times 02 - 17$

12 $01 = 2 \times 43 - 4 \times 17 - 17$

13 $01 = 2 \times 43 - 5 \times 17$

14 So, $a^{-1} = -5[43] = 38$

Modular Exponentiation

Modular exponentiation is a type of exponentiation performed over a modulus $m \in \mathbb{Z}^+$. I.e., solving formula of the form (where $a, b \in \mathbb{Z}$):

$$a^b[m]$$

We can use the naive approach (inefficient for large b or m), repeated squaring (efficient when b is power of 2), and the Fermat little theorem, which says: If $m \in \mathbb{Z}^*$ is prime and $\gcd(a, m) = 1$, then $a^{m-1} \equiv 1[m]$.

Example. Let's compute the following: $\Sigma = 3^{16384}[23] + 61^{730}[53]$.

Modular exponentiation is the foundation of many cryptographic algorithms: RSA, El Gamal, DHKE, DSA, Schnorr signature, zero-knowledge proofs, ECC-based protocols (e.g., ECDH), etc.

Fun Fact. When you type in <https://www.whatever.something> on your web-browser and hit enter, multiple modular exponentiation operations take place (your browser and the remote server compute them), so that you establish a secure connection with the remote server.

Modular Arithmetic Hard Problems

Crypto uses modular arithmetic to leverage the following hard problems (easy to compute, hard to reverse — **computationally** infeasible):

- **Discrete Logarithm Problem.** Given $g \in \mathbb{Z}^+$ and $a \in \mathbb{Z}^+$, it is easy to compute $g^a[m]$. However, given a large x , finding $a \geq 0$ such that $x \equiv g^a[m]$ is a difficult problem to solve.

Let m be a prime number, $g \in \mathbb{Z}^+$ a positive integer, and $x \in \mathbb{Z}_m^*$. If a can be computed using current computing power, such that $x \equiv g^a[m]$, then DHKE and DSA are broken (Also, when m and g are badly chosen).

- **Prime Factorization Problem.** Every integer (greater than 1) can uniquely be factored into a product of two or more prime numbers (FTA). Given k prime numbers $p_1, p_2, \dots, p_k \in \mathbb{Z}^+$, it is easy to compute the composite number $n = (p_1 \cdot p_2 \cdot \dots \cdot p_k)[m]$. However, it is hard to find a specific factorization of the composite n .

Let p and q be very large prime numbers, and $n = (p \cdot q)[m]$. If n can be factored using current computing power, then RSA is broken.

Classical (Historical) Cryptography

Classical Cryptography — Historical Encryption

Definition

Classical Cryptography. Was a form of cryptography (i.e., uses math to encrypt text) that was used until the early to mid-20th century.

It relied on the the following:

- Used only to protect text (e.g., written documents).



- They are less secure by modern standards.
- Many classical ciphers did not adhere to the Kerckhoff's doctrine.
- Was based on substitution, permutation, and transposition.
- “Unawarengly” adopted modular arithmetic.

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It is **Substitution.** In this method, the units of text are replaced with other units of text, according to a fixed system, to encrypt the text. The units may be constituted of single letter (monoalphabetical), or multiple letters (polyalphabetical). The receiver decrypts the text by performing the inverse substitution. E.g.,

- Plaintext: HELLO.
- Ciphertext: KHOOR.
- Key: Each letter in the plaintext is shifted by three positions to the right ($H \ggg K$, $E \ggg H$, $L \ggg O$, $L \ggg O$, and $O \ggg R$).

• was based on substitution, transposition, and permutation.

Classical Cryptography — Historical Encryption

Definition

Classical Cryptography. Was a form of cryptography (i.e., uses math to encrypt text) that was used until the early to mid-20th century.

It is **Transposition.** In this method, the units of the plaintext are rearranged in a different order, but the units themselves are left unchanged. Hence, the method performs permutation between units of the plaintext. The receiver performs the inverse transposition.

- Plaintext: HELLO.
- Ciphertext: HOLEL.
- Key: The plaintext is arranged in zigzag pattern with three rows. Letters are then read row by row following zigzag pattern.

- Was based on substitution, transposition, and permutation.

Caesar Cipher

Substitution-based Ciphers (Caesar Cipher)

Caesar Cipher (Julius Caesar — 100-40AD): is a mono-alphabetical substitution cipher. It relies on a cyclic shifting of letters with respect to a secret key $k=3$ (i.e., shift each letter three positions to the right when encrypting, and three positions to the left when decrypting).



Substitution-based Ciphers (Caesar Cipher)

Encryption: Having a plaintext $X = (x_1, \dots, x_n)$ and a secret key k (a letter), apply the key k on the plaintext X as follows ($n \in \mathbb{N}$):

$$E_k(X) = \begin{cases} (x_1, \dots, x_n), & \text{if } \sigma(k) = 0, \\ Y = (y_1, \dots, y_n), & \text{otherwise.} \end{cases}$$

Where: $y_1 = \sigma^{-1}((\sigma(x_1) + \sigma(k))[26]), \dots, y_n = \sigma^{-1}((\sigma(x_n) + \sigma(k))[26])$.

Decryption: Having a ciphertext $Y = (y_1, \dots, y_n)$ and a secret key k , apply the key k on the ciphertext Y as follows ($n \in \mathbb{N}$):

$$D_k(Y) = \begin{cases} (y_1, \dots, y_n) & \text{if } \sigma(k) = 0, \\ X' = (x'_1, \dots, x'_n), & \text{otherwise.} \end{cases}$$

Where: $x'_1 = \sigma^{-1}((\sigma(y_1) - \sigma(k))[26]), \dots, x'_n = \sigma^{-1}((\sigma(y_n) - \sigma(k))[26])$.

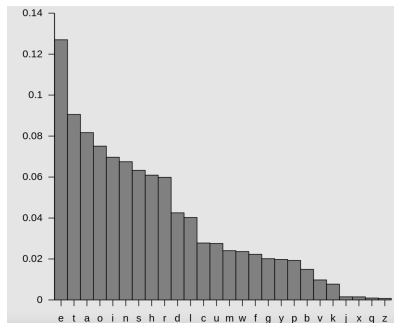
Let $\mathcal{A} = \{a, \dots, z\}$ be the set of alphabets. Then, $\sigma : \mathcal{A} \rightarrow [0 - 25]$ is a character valuation function. The value $\sigma(x_i | y_i) \in [0 - 25] (i \in \mathbb{N}^*)$ is the valuation of a given alphabetical letter. We should have $X = X'$

Substitution-based Ciphers (Caesar Cipher)

Cryptanalysis: The cipher is a monoalphabetical substitution scheme. I.e., the same letter is encrypted with the same code in the plaintext:

Two possible attack patterns:

- ① Brute force attack can be employed to crack the cipher by trying all possible keys. There is a total of **???** possible keys.
- ② Vulnerable to frequency analysis attacks (E.g., Al-Kindi attack), where we consider the most used letter, bigrams (e.g., es, de, le, ...), and trigrams (e.g., the, and, les, ...).



Try cracking this message: Z**o**b**c****f****o****b****k****x****m****o** z**k****i****c** y**p****p**

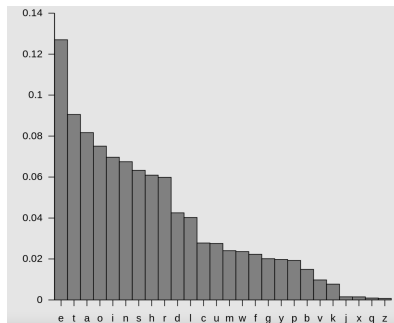
You may use this service: [Decode Caesar](#).

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- ② Vulnerable to frequency analysis attacks (E.g., Al-Kindi attack), where we consider the most used letter, bigrams (e.g., es, de, le, ...), and trigrams (e.g., the, and, les, ...).



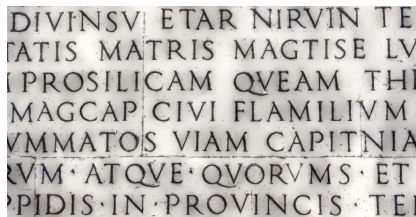
Try cracking this message: Z**o**b**c****f****o****b****k****x****m****o** z**k****i****c** y**p**p

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Affine Cipher

Substitution-based Ciphers (Affine Cipher)

Affine Cipher (Unknown author): is a mono-alphabetical substitution cipher, where each letter is mapped into its numeric value, multiplied by a coefficient (a), and then shifted by another constant (b).



It is a generic form of the Caesar cipher ($a=1$, and b is the shifting key).

Substitution-based Ciphers (Affine Cipher)

Encryption: Having a plaintext $X = (x_1, \dots, x_n)$ and a secret key $k = (a \neq 0, b)$ (two valuated letters), apply a linear function ($A(x) = ax + b$) on the plaintext X as follows (where, $n \in \mathbb{N}$):

$$E_k(X) = Y = (a \cdot x_i + b)[26], \quad \forall x_i \in X$$

Decryption: Having a ciphertext $Y = (y_1, \dots, y_n)$ and a secret key k , apply the key k on the ciphertext Y as follows ($n \in \mathbb{N}$):

$$D_k(Y) = (a^{-1} \cdot (y_i - b))[26], \quad \forall y_i \in Y$$

Where a^{-1} is the inverse of a modulo 26. I.e., a and the size of alphabet $m = 26$ **must be coprime** (i.e., $\gcd(a, m)=1$):

Why?

Substitution-based Ciphers (Affine Cipher)

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Where a^{-1} is the inverse of a modulo 26. I.e., $a \in \mathbb{N}^*$ and the size of alphabet $m = 26$ **must be coprime** (i.e., $\gcd(a, m) = 1$):

- $E_k(\cdot)$ and $D_k(\cdot)$ are injective, so one plaintext maps to one unique ciphertext, and vice-versa (a kind of bijection).
- Ensures the existence of the inverse [26] of the parameter a .

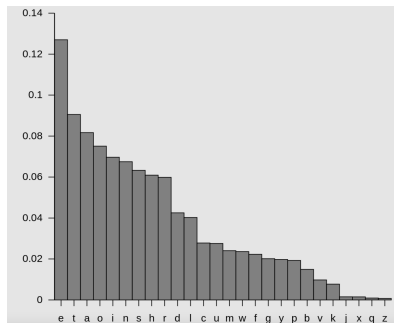
$$\exists a^{-1} \in [1 - 26] : a \cdot a^{-1} \equiv 1[26]$$

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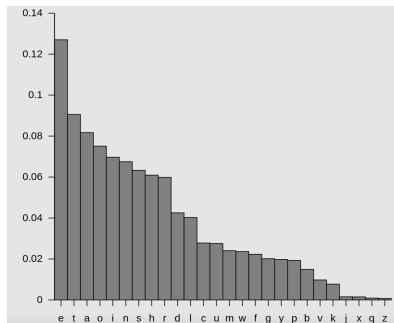
You may use this service: [Decode Affine](#).

Substitution-based Ciphers (Affine Cipher)

Cryptanalysis: The cipher is a monoalphabetical substitution scheme. I.e., the same letter is encrypted with the same code in the plaintext:

Two possible attack patterns:

- ① Brute force attack can be employed to crack the cipher by trying all possible keys. There is a total of **312** possible keys.
- ② Vulnerable to frequency analysis attacks (E.g., Al-Kindi attack), where we consider the most used letter, bigrams (e.g., es, de, le, ...), and trigrams (e.g., the, and, les, ...).



Try cracking this message: Apcdpgpclynp aljd zqq

You may use this service: [Decode Affine](#).

Vigenère Cipher

Substitution-based Ciphers (Vigenère Cipher)

Vigenère Cipher (Blaise de Vigenère — 16th centry): The Vigenère cipher is a poly-alphabetical substitution cipher. It is a variante of Caesar cipher in which the secrete key k is a keyword and not a single letter.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Substitution-based Ciphers (Vigenère Cipher)

Encryption: Given a plaintext $X = (x_1, \dots, x_n, \dots, x_{m-n+1}, \dots, x_m)$ and a secret keyword $K = (k_1, \dots, k_n)$, apply the key k on the plaintext X as follows, where $n, m \in \mathbb{N}$, $m \geq n$, $\mathcal{A} = \{a, \dots, z\}$, and $\sigma : \mathcal{A} \rightarrow [0-25]$ the character valuation function ($\sigma^{-1}(\cdot)$ its inverse function):

$$E_k(X) = (E_k(x_1, \dots, x_n), \dots, E_k(x_{m-n}, \dots, x_m)) = \\ (E_k(x_1, \dots, x_n) = (\sigma(x_1) + \sigma(k_1), \dots, \sigma(x_n) + \sigma(k_n)) [26],$$

$$\vdots,$$

$$E_k(x_{m-n}, \dots, x_m) = (\sigma(x_{m-n}) + \sigma(k_1), \dots, \sigma(x_m) + \sigma(k_n)) [26])$$

Decryption: Having a ciphertext $Y = (y_1, \dots, y_n, \dots, y_{m-n}, \dots, y_m)$ and the secret keyword $K = (k_1, \dots, k_n)$, do the following:

$$D_k(Y) = (D_k(y_1, \dots, y_n), \dots, D_k(y_{m-n}, \dots, y_m)) = \\ (D_k(y_1, \dots, y_n) = (\sigma^{-1}((\sigma(y_1) - \sigma(k_1))[26]), \dots),$$

$$\vdots,$$

$$D_k(y_{m-n}, \dots, y_m) = (\sigma^{-1}((\sigma(y_{m-n}) - \sigma(k_1))[26]), \dots))$$

Substitution-based Ciphers (Vigenère Cipher)

Cryptanalysis: The cipher relies on polyalphabetic substitution, i.e. the same letter may not be ciphered with the same code within the same plaintext. Thus, it is more resistant to frequency analysis attacks.

Example. Use the Vigenère table and the cryptographic key $k = \text{ensia}$ to decrypt the following ciphertexts:

- ① Scwzaxvfo Scflmmw
- ② Rrleovxk inh Cjwtspgts
- ③ Gbexuxrj inh Awbwsec Aeghjqtq

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Substitution-based Ciphers (Vigenère Cipher)

Cryptanalysis: The cipher relies on polyalphabetic substitution, i.e. the same letter is not ciphered with the same code within the same plaintext. Thus, it is more resistant to frequency analysis attacks.

Example. Use the Vigenère table and the cryptographic key $k = \text{ensia}$ to decrypt the following ciphertexts:

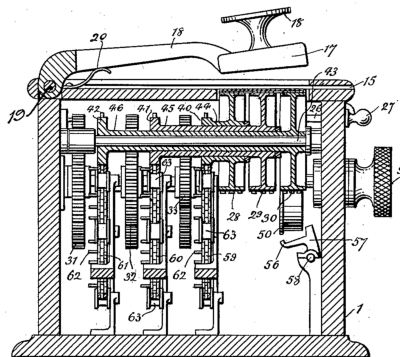
- ① Operating Systems
- ② Networks and Protocols
- ③ Computer and Network Security

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Hill Cipher

Substitution-based Ciphers (Hill Cipher)

Hill Cipher (Lester S. Hill — 1929): It is a polygraphic substitution cipher that employs matrices to encrypt text through matrix multiplication over finite field. The key in this cipher is a matrix.



Substitution-based Ciphers (Hill Cipher)

Encryption: Having a plaintext $X = (x_1, \dots, x_n)$ and a secret square matrix k of degree $m \in \mathbb{N}$, apply the key k on the plaintext X as follows:

$$\begin{aligned} E_k(X) &= (E_k(x_1, \dots, x_m), \dots, E_k(x_{n-m+1}, \dots, x_n)) \\ &= (k \times ((\sigma(x_1), \dots, \sigma(x_m))) [26], \dots, (k \times (\sigma(x_{n-m+1}), \dots, \sigma(x_n))) [26]) \end{aligned}$$

Where $k = \begin{pmatrix} k_{1,1} & \dots & k_{1,m} \\ \vdots & \dots & \vdots \\ k_{m,1} & \dots & k_{m,m} \end{pmatrix}$, $\det(k) \neq 0$ (i.e., invertible).

Decryption: Having a ciphertext $Y = (x_1, \dots, x_n)$ and a secret square matrix k of degree $m \in \mathbb{N}$, apply the key k on the ciphertext Y as follows:

$$\begin{aligned} D_k(X) &= (D_k(y_1, \dots, y_m), \dots, D_k(y_{n-m+1}, \dots, y_n)) \\ &= (\sigma^{-1}(k^{-1} \times (\sigma(y_1), \dots, \sigma(y_m))) [26], \dots) \end{aligned}$$

Where $k^{-1} \times k \equiv k \times k^{-1} \equiv \begin{pmatrix} 1 & \dots & 1 \\ \vdots & 1 & \vdots \\ 1 & \dots & 1 \end{pmatrix} [26]$.

Substitution-based ciphers (Hill Cipher)

Example: Consider the following Hill cipher with key k (given below). Encrypt the word “MATH”.

$$k = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$$

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$$k = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$$

The order of the matrix is $m=2$. Hence, we divide the plaintext into blocks of 2 letters. We have, for the first block, $MA = (\sigma(M), \sigma(A)) = (12, 0)$ and for the second one $TH = (\sigma(T), \sigma(H)) = (19, 7)$.

The product gives: $(12, 0) \times k = (4, 8)$, and $(19, 7) \times k = (17, 14)$

The ciphertext is: $(\sigma^{-1}(4), \sigma^{-1}(8))$ and $(\sigma^{-1}(17), \sigma^{-1}(14)) \equiv \text{EIRO}$

Decrypt the ciphertext “EIRO” and recover the original message.

Cryptanalysis: ?

Substitution-based ciphers (Hill Cipher)

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Decrypt the ciphertext “EIRO” and recover the original message.

Cryptanalysis: Because it is completely linear, the Hill cipher is vulnerable to a known-plaintext attack. Also, brute-forcing the cipher consists of testing all square matrices that are invertible mod 26.

Substitution-based ciphers (Hill Cipher)

Example: Consider the following Hill cipher with key k (given below). Encrypt the word “MATH”.

$$k = \begin{pmatrix} 9 & 4 \\ - & - \end{pmatrix}$$

Classical cryptographic attacks:

- **Known-Plaintext attack.** The attacker has access to both the plaintext and the ciphertext and tries to deduce the key.
- **Known-Ciphertext attack.** The attacker has access to the ciphertext and uses statistical analysis to deduce the key.
- **Chosen-Plaintext attack.** The attacker use plaintexts to get their ciphertexts, to use the first attack above.
- **Chosen-Ciphertext attack.** The attacker use ciphertexts to get their plaintexts, to use the first attack above.

of testing all square matrices that are invertible mod 26.

Rail Fence Cipher

Transposition-based Ciphers (Rail Fence Cipher)

Rail Fence Cipher (Spartans 404 AD): Is a transposition cipher that uses a table that looks a bit like an old rail fence viewed from above.



Transposition-based Ciphers (Rail Fence Cipher)

Encryption: Write the plaintext (e.g. **we need help**) on a table of k rows (e.g., $k = 3$) s.t. the plaintext appears in a zigzag form (rail fence). The ciphertext (e.g., **weleedepnh**) is then read row by row.

W				E				L	
	E		E		D		E		P
		N				H			

Decryption: Write the cyphertext (e.g. **weleedepnh**) row by row in a table that has k rows (e.g. $k = 3$) and a number of columns equal to the size of the cyphertext (e.g. 10) considering the zigzag form (rail fence) of the text.

W				E				L	
	-		-		-		-		-
		-				-			

Transposition-based Ciphers

Transposition-based Ciphers

Block-Transposition Cipher: This cipher deals with blocks. It rearranges the characters within a block according to a bijective mapping (e.g., a permutation matrix π).

$$\pi = \begin{pmatrix} 1 & \dots & n \\ \pi(1) & \dots & \pi(n) \end{pmatrix}$$

Encryption. To encrypt, the text is divided into equal sized blocks. Then, within each bloc, the letters are permuted based on π .

$$E_{\pi}(X) = (x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(n)})$$

E.g.,

$$\pi = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

The plaintext “I WANT” becomes “WAI T*N”.

Transposition-based Ciphers

Decryption. To decrypt, the ciphertext is divided into equal sized blocks. Then, within each block, the letters are permuted based on π' .

$$E_{\pi'}(X) = (x_{\pi'(1)}, x_{\pi'(2)}, \dots, x_{\pi'(n)})$$

E.g., Here π' is the inverse of π :

$$\pi' = \pi^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

The ciphertext “WAI T*N” becomes “I WANT”.

Cryptanalysis. This cipher is more resilient than the previous ones. For a matrix π of n columns, the number of possible matrices is ???.

Transposition-based Ciphers

Decryption. To decrypt, the ciphertext is divided into equal sized blocks. Then, within each block, the letters are permuted based on π' .

$$E_{\pi'}(X) = (x_{\pi'(1)}, x_{\pi'(2)}, \dots, x_{\pi'(n)})$$

E.g., Here π' is the inverse of π :

$$\pi' = \pi^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

The ciphertext “WAI T*N” becomes “I WANT”.

Cryptanalysis. This cipher is more resilient than the previous ones. For a matrix π of n columns, the number of possible matrices is $n!$.

- **End.**